Supplementary Materials for Manuscript Entitled

Taxonomic and functional dysregulation in salivary microbiomes during oral carcinogenesis

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Running Head: Oral Microbiome in Malignant Transformation of OVH

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Table S1. Descriptive data of the study cohort. The former drinker/smoker/betel nut chewer indicates those who had quit drinking, smoking, or betel nut at the time of interview. The current one indicates those who still have the habits occasionally or regularly.

Cohort	Sex	Age	Alcohol	Betel nut	Cigarette	Cancer stage
Normal (n=27)	Male=27	47.4±9.6	never=14 former=1 current=12	never=9 former=4 current=14	never=8 former=4 current=15	N/A
OVH (n=21)	Male=21	51.0±13.2	never=6 former=1 current=14	never=1 former=5 current=15	never=2 former=2 current=17	N/A
OSCC (n=27)	Male=27	54.4±10.1	never=12 former =3 current=12	never=3 former=21 current=3	never=3 former=8 current=16	I = 12 II = 6 III = 2 IV= 7
Overall (n=75)	Male=75	50.9±11.2	never=32 former=5 current=38	never=13 former=30 current=32	never=13 former=14 current=48	I = 12 II = 6 III = 2 IV= 7

* N/A: not applicable

Table S2. Pairwise comparison of cohort metadata using the univariate analysis.

Group 1	Group 2	Age		Alcohol		Betel nut		Cigarette	
		t-statistic	p-value	H-statistic	p-value	H-statistic	p-value	H-statistic	p-value
Normal (n=27)	OVH (n=21)	1.117	0.270	2.521	0.112	3.265	0.071	3.6	0.058
Normal (n=27)	OSCC (n=27)	2.624	0.011	0.084	0.772	1.254	0.263	0.59	0.442
OVH (n=21)	OSCC (n=27)	0.996	0.325	1.973	0.16	15.468	8.39E-05	2.073	0.150

Table S3. Pairwise comparison of alpha diversity indices using the Kruskal-Wallis test.

Group 1	Group 2	Pielou's evenness			Observed ASVs			Shannon's index			Faith's phylogenetic diversity		
		H	p-value	q-value	H	p-value	q-value	H	p-value	q-value	H	p-value	q-value
Normal (n=27)	OVH (n=21)	0.0390	0.843	0.843	4.065	0.044	0.131	1.058	0.304	0.455	1.190	0.275	0.485
Normal (n=27)	OSCC (n=27)	2.0867	0.149	0.446	2.425	0.119	0.179	3.144	0.076	0.229	0.081	0.775	0.775
OVH (n=21)	OSCC (n=27)	0.7085	0.400	0.600	0.691	0.406	0.406	0.048	0.827	0.827	0.975	0.324	0.485

q-value is the FDR-adjusted p-value for multiple comparisons.

Table S4. Signature species with relative abundance (Mean ± SD, %) in the studied cohorts.

Species	Normal	OVH	OSCC
Parvimonas micra	0.13±0.13	0.26±0.75	0.33±0.31
Actinomyces sp. HMT-180	0.86±0.76	0.10±0.16	0.18±0.40
Peptostreptococcus stomatis	0.14±0.19	0.31±0.86	0.90±1.15
Rothia mucilaginosa	1.40±1.87	0.52±0.76	0.63±1.62
Prevotella salivae	1.70±1.42	0.65±0.73	1.34±1.78
Rothia dentocariosa	0.97±1.85	1.47±2.53	0.48±0.95
Leptotrichia sp. HMT-417	1.18±1.85	0.45±0.73	0.25±0.55
Campylobacter concisus	1.79±1.27	0.60±0.86	0.75±1.04
Megasphaera micronuciformis	1.68±1.70	0.48±0.56	0.51±0.81
Selenomonas unclassified*	1.50±1.30	1.49±1.87	0.59±0.46
Prevotella oris	0.57±0.62	0.91±0.83	1.98±2.27
Veillonella parvula	1.35±1.56	2.41±3.67	0.82±1.96
Prevotella pallens	1.90±3.60	0.61±1.08	1.52±3.92
Veillonella dispar	3.62±3.33	1.76±2.12	1.67±2.56
Saccharibacteria (TM7) [G-1] bacterium HMT-352	2.98±3.57	0.99±2.35	0.89±2.63
Haemophilus parainfluenzae	2.85±3.16	1.15±2.37	1.51±2.34
Streptococcus salivarius	2.40±4.58	0.88±1.76	0.22±0.36
Prevotella unclassified*	3.49±4.60	1.54±2.74	0.63±0.79
Veillonella atypica	3.99±4.13	3.04±5.75	1.15±2.10
Prevotella melaninogenica	4.75±5.61	1.40±1.89	2.20±3.70
Capnocytophaga sputigena	0.24±0.31	0.70±1.15	4.69±9.49

The asterisk (*) indicates a taxon that was annotated only to the genus level.

Table S5. Studies related to the oral microbiome and oral cancer since 2015. Only studies with samples collected by non-invasive collection methods (oral swab, oral rinse, or saliva samples) was used in the present study. Data sets were downloaded on April, 2020

Ref.	Year	Sample type	Platform	HVR	Accession number	Comment	Used in this study
[1]	2015	tissue	Roche 454	V1-V3	SRA204252	Not Illumina platform, tissue sample	No
[2]	2016	tissue and saliva	Roche 454	V3-V5	NA	-	No
[3]	2016	saliva	Illumina MiSeq	NA	NA	-	No
[4]	2017	saliva	Illumina TruSeq	V4	SRP107079	Merged sequences, not suitable for DADA2 pipeline	No
[5]	2017	saliva and swab	Roche 454	V3-V5	SAMN06127413	Not Illumina platform, no sample barcodes	No
[6]	2017	oral rinse	Illumina MiSeq	V4	PRJNA321193	Dubious sequences	No
[7]	2017	saliva	Illumina MiSeq	V4	PRJEB18476	Illumina platform, saliva samples	Yes
[8]	2017	swab	Illumina MiSeq	V4-V5	SRP097643	Illumina platform, oral swab samples	Yes
[9]	2017	tissue	Illumina MiSeq	V1-V3	PRJNA352375	Tissue sample	No
[10]	2017	swab	Illumina MiSeq	V1-V2	PRJNA394711	Healthy and precancer cohort, no cancer cohort	No
[11]	2017	saliva	Roche 454	V3-V5	NA	-	No
[12]	2017	tissue	Ion Torrent	V4	Upon request	Not Illumina platform, tissue sample	No
[13]	2017	tissue	Clone	V1-V4	LT674587-LT699699	Not Illumina platform, tissue sample	No
[14]	2018	oral rinse	Illumina MiSeq	V6-V8	NA	-	No
[15]	2018	tissue	Illumina MiSeq	V1-V3	PRJNA415963	tissue sample	No
[16]	2018	oral rinse	Illumina MiSeq	V3-V4	NA	-	No
[17]	2018	saliva	Illumina MiSeq	V4	NA	-	No
[18]	2018	oral rinse	Roche 454	V3-V4	NA	-	No

[19]	2018	saliva	Illumina MiSeq	V3-V5	NA	-	No
[20]	2019	saliva	Illumina MiSeq	V1-V2	NA	-	No
[21]	2019	saliva	Illumina MiSeq	V4	NA	-	No
[22]	2019	oral rinse	Roche 454	V3-V4	NA	-	No
[23]	2020	swab	Illumina MiSeq	V3-V4	PRJNA533177	No data found	No

Table S6 Multiple linear regression of the functionality and familty taxa detected in the oral microbiomes.

	Cofactor, Prosthetic Group, Electron Carrier, and Vitamin Biosynthesis		
Coefficients	Normal	NRVH	OSCC
X0 (constant)	14.846	13.9536	11.2937
X1 (Actinomycetaceae)	**-0.1425**	**-1.1287**	**-0.7065**
X2 (Flavobacteriaceae)	-0.0556	0.0732	0.0342
X3 (Peptostreptococcaceae)	-0.0301	-0.006	**0.442**
X4 (Veillonellaceae)	0.0119	0.0148	**0.1761**
R-squared	0.047	0.259	0.339
adjusted R-squared	-0.126	0.074	0.219

	Nucleoside and Nucleotide Biosynthesis		
Coefficients	Normal	NRVH	OSCC
X0 (constant)	13.793	13.1739	16.4389
X1 (Actinomycetaceae)	**0.1153**	**0.3727**	**1.0284**
X2 (Flavobacteriaceae)	**0.125**	-0.0116	-0.041
X3 (Peptostreptococcaceae)	**-0.3399**	**0.3314**	**-0.4029**
X4 (Veillonellaceae)	-0.0053	0.0037	**-0.1562**
R-squared	0.271	0.343	0.389
adjusted R-squared	0.138	0.179	0.278

	Cell Structure Biosynthesis		
Coefficients	Normal	NRVH	OSCC
X0 (constant)	6.7241	6.6351	7.9722
X1 (Actinomycetaceae)	0.0549	**0.4687**	**0.4458**
X2 (Flavobacteriaceae)	0.0388	-0.0242	-0.0304
X3 (Peptostreptococcaceae)	**-0.1245**	0.037	**-0.1709**
X4 (Veillonellaceae)	0.0194	0.0131	-0.0518
R-squared	0.438	0.341	0.485
adjusted R-squared	0.336	0.177	0.391

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